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🔗 Blockchain-Based Blue Carbon Registry and MRV System 🌱

- **Problem Statement ID** – 25038.
- **Problem Statement Title** – Blockchain-Based Blue Carbon Registry and MRV System.
- **Theme** – Clean & Green Technology.
- **PS Category** – Software.
- **Team Name** – HEXA VISION

HEXA VISION

Blockchain-Based Blue Carbon Registry & MRV Platform

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❖ Problems :

⚠️ **Lack of transparency** – in carbon credit tracking

⏳ **Slow verification** – for blue carbon projects (mangroves , seagrass , salt marshes)

🌱 **Difficult for authorities** – to authenticate local initiatives

❖ Proposed Solution / Idea :

🔗 **Blockchain platform** – for blue carbon credit tracking

🤖 **Smart contracts** – Automatic tokenization

📱 **Mobile app + Drones + IoT** – real-time data capture

📊 **Admin dashboard** – for authority verification

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TECHNICAL APPROACH

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Technologies :

Blockchain : Hyperledger or Ethereum
Smart Contracts : Solidity
Mobile App : Flutter / React Native
Dashboard : React + Node.js
Cloud DB : AWS / Azure
IoT

Methodology Flow :

Data Capture
Smart Contract Validation
Token Generation
Authority Verification
Carbon Credit Issuance

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FEASIBILITY AND VIABILITY

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Feasibility ✅ :	Impact 🌍 :
• Proven tech: Blockchain + IoT • Scalable across coastal states • Offline data sync • Govt partnerships & phased rollout • Open-source tools to reduce cost	• 🌱 Environmental : Protect mangroves & seagrass • 💼 Economic : Credible carbon offset for industries • 👥 Social : Jobs for local communities

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IMPACT AND BENEFITS

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Environmental 🌱 :

- Preserve mangroves, seagrass, and coastal habitats
- Increase carbon sequestration and combat climate change

Economic 💼 :

- Enable verified carbon credits for industries
- Promote sustainable investments and green economy

Social 👥

- Generate livelihood opportunities for local communities
- Raise climate awareness and community participation

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RESEARCH AND REFERENCES

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- **UN Blue Carbon Reports** – Global guidelines on coastal ecosystems
- **Govt of India – MoEFCC Guidelines** – National framework for carbon credits
- **World Bank – Blockchain in Carbon Markets** – Use of emerging tech for climate action
- **Key Research Papers** – Blue carbon restoration & verification models

IMPORTANT INSTRUCTIONS

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Please ensure below pointers are met while submitting the Idea PPT:

1. **Kindly keep the maximum slides limit up to six (6). (Including the title slide)**
2. **Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures**
3. **Keep your explanation precise and easy to understand**
4. **Idea should be unique and novel.**
5. **You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).**
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